PROJECT SYNOPSIS

CORTEX HUB

Enterprise AI Component Store & Customization Platform

Project Type	Web-based AI Marketplace and SaaS Platform
Domain	Artificial Intelligence, Machine Learning, Enterprise Software
Repository	Goldypahal/Cortex-Hub
Technology	HTML5, Modular JavaScript, CSS, Node.js
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Submitted by: _____

Department / Institution: _____

A synopsis describing the purpose, scope, methodology, architecture, modules, and expected outcomes of the Cortex Hub project.

1. Abstract

Cortex Hub is an enterprise-focused AI component marketplace and customization platform designed to help organizations discover, evaluate, purchase, customize, and deploy production-ready artificial intelligence components. The platform goes beyond basic chatbot demonstrations by presenting reusable AI engines such as Retrieval-Augmented Generation (RAG) chatbots, customer-support agents, document intelligence systems, resume matching engines, legal analysis tools, financial report analyzers, SQL agents, code reviewers, multi-agent workflow systems, and OCR-based extraction solutions.

The proposed system combines a customer-facing marketplace, an interactive AI playground, an architecture and Bill of Materials (BOM) configurator, a customer portal, an administration dashboard, licensing and checkout workflows, and an underlying AI services layer. The platform emphasizes grounded responses, exact source citations, hybrid retrieval, usage monitoring, tenant isolation, and deployment flexibility including cloud, private Virtual Private Cloud (VPC), on-premises, and air-gapped environments.

Cortex Hub is intended to reduce the time and technical effort required for organizations to adopt AI. Instead of developing every AI solution from the beginning, a business can select a verified component, test it with representative data, configure it for its domain, receive an estimated cost and delivery timeline, and manage the resulting licensed solution through a centralized portal.

2. Introduction

Artificial intelligence adoption is increasing across healthcare, finance, legal services, education, human resources, and other enterprise domains. However, organizations often face difficulties in selecting reliable AI components, validating their performance, integrating them with existing data, controlling inference costs, and deploying them securely. Generic AI demos may demonstrate a model's capabilities but frequently lack production-oriented features such as citation grounding, access control, auditability, licensing, monitoring, and customization workflows.

Cortex Hub addresses this gap by providing a structured store of reusable AI components and a guided path from discovery to deployment. The project treats AI functionality as configurable software products rather than isolated model demonstrations. It also introduces a human-in-the-loop customization process so that complex enterprise requirements can be reviewed, quoted, implemented, and tracked.

3. Problem Statement

Businesses that want to adopt AI frequently need to compare multiple solutions, understand their technical architecture, test them on sample documents, estimate deployment costs, and request modifications. These activities are commonly fragmented across vendor websites, technical consultations, disconnected demos, and manual communication. As a result, organizations may experience high evaluation costs, uncertain delivery timelines, poor visibility into usage, and difficulty maintaining security and compliance.

There is a need for a unified platform that presents reusable AI components with technical transparency, supports controlled experimentation, provides configurable deployment options, and manages the commercial and operational lifecycle of each component.

4. Objectives

- Develop a centralized marketplace for discovering and comparing production-ready AI components.
- Provide an interactive sandbox where users can test AI components using sample documents and predefined scenarios.

- Support grounded AI responses with source citations, confidence indicators, latency, token usage, and estimated cost.
- Create a guided customization configurator covering domain, data sources, authentication, model provider, deployment, branding, and feature options.
- Generate estimated pricing and delivery timelines through a rule-based BOM and estimation engine.
- Provide secure checkout, licensing, invoices, and maintenance or hosting subscription workflows.
- Offer a tenant-aware customer portal for API keys, usage metrics, documentation, purchases, and support.
- Provide an admin operations desk for catalog management, customization requests, deployments, customers, and revenue analytics.
- Design the platform to support cloud, private VPC, on-premises, and air-gapped deployment models.

5. Scope of the Project

In Scope

- Public marketplace with catalog browsing, search, category and industry filters, product pages, pricing, architecture diagrams, and SDK examples.
- Interactive playground supporting temporary PDF/DOCX uploads, model and retrieval selection, cited answers, and telemetry.
- Customization wizard and request lifecycle from submission through review, quotation, implementation, and delivery.
- Checkout, tax calculation, invoices, license records, renewal status, and simulated payment methods.
- Customer dashboard with organization details, API-key management, usage analytics, documentation, support tickets, and deployment status.
- Admin dashboard with role-based access, catalog CRUD, request Kanban, customer management, revenue analytics, and deployment tracking.
- AI services concepts including hybrid dense and lexical retrieval, reranking, citation generation, model routing, evaluation, and usage tracking.

Out of Scope for the Initial Version

- A fully autonomous customization process without human review.
- A third-party developer marketplace for publishing external components.
- Native Android or iOS applications.
- A complete global tax and multi-currency engine.
- Unrestricted free inference or unlimited playground usage.

6. Proposed System

The proposed system is organized into three interconnected planes. The customer-facing plane manages discovery, demos, configuration, purchase, and customer self-service. The admin plane manages products, customers, requests, billing-related information, support, analytics, and deployments. The AI services plane provides the intelligence used by the playground and deployed components.

Plane	Main Responsibilities

Customer-Facing Plane	Marketplace catalog, live sandbox, architecture configurator, checkout, licensing, and customer portal.
Admin Plane	Customization Kanban, scope review, catalog management, customer management, revenue analytics, and deployment matrix.
AI Services Plane	Document ingestion, hybrid retrieval, reranking, citation engine, model routing, evaluation, tenant isolation, and telemetry.

7. Major Modules

7.1 Marketplace and Product Catalog

Displays AI components with descriptions, categories, industry tags, pricing tiers, architecture diagrams, version information, changelogs, live demo actions, purchase options, and customization actions.

7.2 Live AI Sandbox

Provides a split-pane environment for testing components. Users can upload temporary PDF/DOCX files, ask questions, inspect cited answers, and observe latency, token counts, confidence, and estimated inference cost. Temporary content is isolated and automatically purged according to retention rules.

7.3 Architecture Configurator and BOM Calculator

Guides users through domain selection, data ingestion, authentication, model provider, deployment target, branding, and feature toggles. The system calculates an estimated price and delivery timeline based on configuration rules.

7.4 Checkout and Licensing

Supports component licenses, customization quotes, recurring maintenance or hosting, tax calculation, invoice generation, and unique license records with status and renewal information.

7.5 Customer Portal

Allows organizations to manage purchases, invoices, licenses, API keys, usage quotas, documentation, support tickets, customization requests, and deployment status.

7.6 Administration Dashboard

Provides role-based access for platform administrators, support agents, and billing personnel. It includes catalog CRUD, request Kanban, customer records, deployment tracking, audit logs, and revenue analytics.

7.7 AI Services Layer

Supports document ingestion, chunking, embeddings, hybrid semantic and keyword retrieval, reranking, model selection, source citations, evaluation, REST APIs, SDK access, usage logging, and configurable quotas.

8. Technology Stack

Layer	Technology / Approach
Frontend	HTML5, modular JavaScript using ES Modules, responsive CSS, custom design tokens
Runtime / Development	Node.js, npm, local development server

Architecture Pattern	Component-driven MVC structure with views, data, and utility modules
AI and Retrieval	RAG pipelines, dense vector retrieval, BM25/lexical retrieval, cross-encoder reranking, model routing
Data and Storage	Relational transactional database, vector database, and object storage for documents
Security	Multi-tenant isolation, role-based access control, API-key lifecycle, rate limiting, audit logging
Deployment	Cloud, private VPC, on-premises, and air-gapped deployment models

9. Methodology

Requirement Analysis

Study enterprise AI adoption challenges and define user roles, functional requirements, business rules, and non-functional requirements.

System Design

Design the three-plane architecture, data entities, module boundaries, navigation flows, security controls, and AI service interfaces.

Frontend Development

Implement the marketplace, product detail pages, playground, configurator, checkout, customer portal, and admin dashboard using modular JavaScript and responsive CSS.

AI Service Design

Define document ingestion, chunking, embedding, hybrid retrieval, reranking, response grounding, citation generation, and usage telemetry.

Integration

Connect user workflows with catalog data, customization requests, licensing records, usage metrics, and deployment status.

Testing and Validation

Perform functional, usability, security, integration, and performance testing. Validate that tenant data is isolated and that playground sessions are rate-limited and purged.

Deployment and Documentation

Prepare local deployment instructions, API/SDK documentation, operational procedures, and future extension plans.

10. Functional Requirements Summary

ID	Requirement Summary
FR-1	Browse, search, filter, publish, version, and manage AI components.
FR-2	Run controlled playground experiments with documents, citations, cost, and latency metrics.

FR-3	Configure components and submit customization requests with estimates and status tracking.
FR-4	Purchase licenses, process quotes, generate invoices, and manage renewals.
FR-5	Manage customer accounts, API keys, usage, documentation, support, and organization members.
FR-6	Provide admin roles, catalog management, request Kanban, analytics, audit logs, and deployments.
FR-7	Provide AI ingestion, hybrid retrieval, model routing, citations, evaluation, APIs, and telemetry.

11. Non-Functional Requirements

- Security: Enforce authentication, authorization, tenant isolation, API-key protection, rate limits, and audit logging.
- Reliability: Handle provider failures, invalid files, failed payments, and interrupted customization workflows gracefully.
- Performance: Provide responsive marketplace navigation and track AI response latency and token usage.
- Scalability: Support horizontal scaling of backend and AI services and allow multiple model providers.
- Usability: Offer a clear, responsive interface for business users, developers, administrators, and support staff.
- Maintainability: Use modular frontend architecture, documented APIs, reusable components, and versioned product definitions.
- Privacy: Avoid retaining playground documents beyond the configured retention period and prevent unauthorized reuse of customer data.
- Compliance: Provide design support for applicable data-protection requirements such as India's DPDP framework and GDPR where relevant.

12. Expected Outcomes

- A functional web-based marketplace for reusable enterprise AI components.
- A working demonstration environment for testing AI components before purchase.
- A guided customization workflow with cost and delivery estimation.
- A structured licensing, checkout, and customer-management experience.
- An admin operations dashboard for managing products, requests, customers, and deployments.
- A modular foundation that can later integrate real payment gateways, authentication providers, cloud infrastructure, and production AI services.
- Improved transparency through citations, architecture information, usage telemetry, and configurable deployment choices.

13. Future Enhancements

- Allow verified third-party developers to publish and sell components.
- Add automated deployment pipelines for AWS, Azure, GCP, private VPC, and on-premises environments.

- Integrate production payment gateways, subscription billing, and global tax rules.
- Introduce advanced model evaluation, A/B testing, observability, and automated regression testing.
- Support additional data sources such as websites, Confluence, S3, Snowflake, and enterprise databases.
- Add multilingual interfaces and multilingual retrieval and generation.
- Provide native mobile applications and enterprise single sign-on integrations.
- Add more autonomous customization assistance while retaining human approval for high-impact changes.

14. Conclusion

Cortex Hub proposes a practical and extensible approach to enterprise AI adoption by combining an AI component marketplace with experimentation, customization, licensing, customer operations, and deployment management. Its architecture separates customer-facing workflows, administrative operations, and AI services, making the system easier to maintain and extend. By emphasizing grounded answers, exact citations, tenant isolation, cost visibility, and deployment flexibility, the project provides a foundation for delivering reliable AI capabilities as configurable business software.

The project can serve both as an academic implementation of a modern AI-enabled SaaS platform and as a foundation for a future commercial product offering reusable AI components to organizations with different technical, regulatory, and operational requirements.

15. References

- Cortex Hub GitHub Repository: https://github.com/Goldypahal/Cortex-Hub
- Project README.md — platform overview, architecture, modules, and technology stack.
- Project capstone.md — Software Requirements Specification for the AI Component Store and Customization Platform.
- OWASP Application Security Verification Standard (ASVS) — security baseline reference.
- IEEE 830-1998 — guidance for software requirements specifications.